

Key management and symmetric cryptography

Learning objectives

- **Explain the concept and importance of cryptographic key management**
- **Describe how symmetric encryption works and where it is used**
- **Identify the features and structure of the AES encryption algorithm**
- **Recognise best practices for generating, storing, and rotating encryption keys**

What is key management?

- **Handling cryptographic keys throughout life cycle**
- **Includes generation, distribution, storage, rotation and deletion**
- **Supports security and encryption strength**

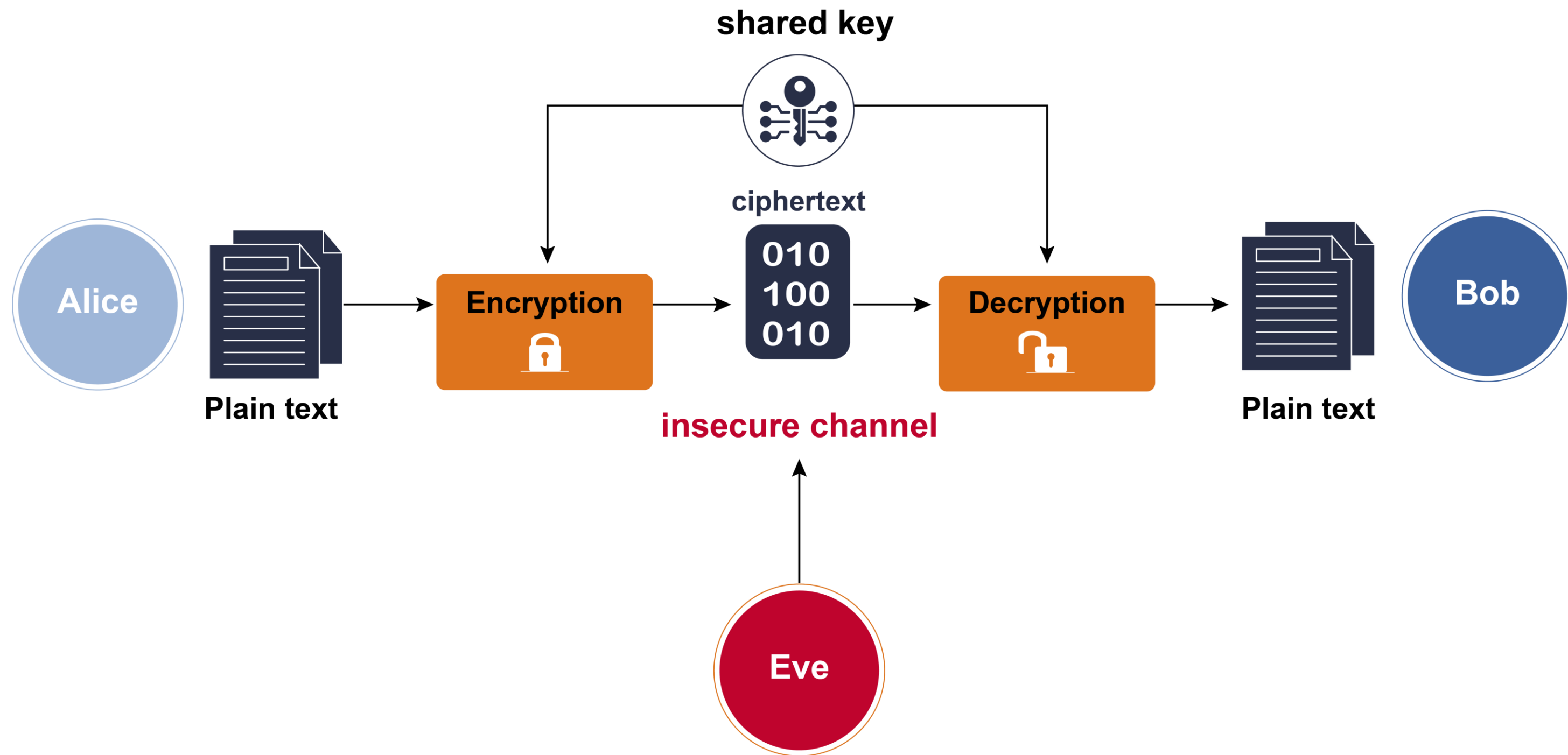
Why key management matters

- **Poor management exposes encrypted data**
- **Strong key policies prevent leaks and misuse**
- **Supports compliance and system trust**

Introduction to symmetric cryptography

- **Same key used for encryption and decryption**
- **Fast and efficient for data processing**
- **Used in secure file storage and apps**

Challenges in symmetric cryptography



What is the Advanced Encryption Standard (AES)?

- **A symmetric encryption standard**
- **Trusted globally by banks and technology companies**
- **Supports 128, 192, 256-bit keys**

How does AES work?

- **Encrypts data in blocks using multiple rounds**
- **Each round makes the data more scrambled and harder to reverse with the correct key**
- **Decryption uses the same key in reverse**

Where symmetric encryption is used



**Database
records**



**Files on
devices**



**Mobile app
communication**



**Virtual private
networks**



**Financial
systems**

Best practices for key management

- **Use secure key generators**
- **Store in protected vaults or hardware**
- **Rotate and restrict key access**

Summary

- **Symmetric encryption is fast and reliable for encrypting data**
- **AES is widely adopted**

Next

- **Asymmetric cryptography and public keys**